

MONISH K

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INTRODUCTION

MS by Research student at IIIT-Bangalore, focusing on graph representation and foundation models. Background in computer science and engineering, interested in applying GRL to NLP and Vision.

EDUCATION

- 1) **MS by Research in Data Science | Deep Learning** (Aug 2024 - Present)
 - 3.92 CGPA out of 4, currently pursuing at IIIT - Bangalore
- 2) **Bachelor of Engineering in Computer Science** (Aug 2018 - Jul 2022)
 - 8.94 CGPA out of 10, from NMIT - Bangalore

RESEARCH EXPERIENCE

- 1) **MS by Research - Graph Representation Learning at IIIT-B** (Aug 2024 - Present)
 - Researching Graph Foundation Models (GFM) for graph-level tasks using self-supervised learning.
 - Developing masked subgraph reconstruction frameworks to study how models infer missing graph regions from boundary information.
 - Theoretical analysis of locality limits in message-passing GNNs and conditions under which global attention becomes necessary.

PROJECTS

- 1) **Multi-Modal Bi-Temporal Remote Sensing Change Detection**
 - Designed a Swin-UNet architecture with novel Differential Window Attention for Change Detection.
 - Demonstrated improved cross-modal generalization compared to baseline change-detection models.
- 2) **Brain Tumor Classification and Segmentation using MRI data - [link](#)**
 - Implemented ViT and 3D Swin-UNETR models for tumor segmentation on BraTS-20 MRI dataset.
 - Integrated both handcrafted and deep features; achieved 89% accuracy on validation set.
- 3) **Smog Intensity Analysis & Forecasting - [link](#)**
 - Extracted spatio-temporal data from satellite using GIS techniques from Google Earth Engine.
 - Applied SARIMA models to forecast smog levels across key highway regions in Karnataka.
- 4) **Image Tamper Detection - [link](#)**
 - Developed a CNN leveraging Error Level Analysis to detect manipulated JPEG images.
 - Achieved 86% accuracy with under 7M parameters; targeted for lightweight forensic use cases.

WORK EXPERIENCE

- 1) **Senior Software Engineer at LTIMindtree** (Nov 2022 - Aug 2023)
 - Engineered complex PL/SQL pipelines for large-scale insurance data reporting and analytics.
 - Led multiple automations to streamline reporting workflows and reduce manual intervention.
- 2) **Graduate Engineer Trainee at LTIMindtree** (Aug 2022 - Nov 2022)
 - Built a full-stack application for a virtual credit card system with controlled e-commerce usage.
 - Focused on backend architecture and transactional flow design using standard MVC principles.
- 3) **Technical Intern at Egnite** (Jun 2021 - Dec 2021)
 - Built web scrapers and led NLP analysis for visualizing ML-related keywords' popularity over time.
 - Managed GitHub tasks, verified code submissions, and ensured project quality.

SKILLS

- 1) **Programming** - Python, C/C++, Java, SQL
- 2) **Research Areas** - Machine Learning, Deep Learning, Graph Representation Learning, NLP, Vision
- 3) **Framework & Tools** - PyTorch, TensorFlow, Scikit-learn, OpenCV, NLTK, NetworkX, PyG, Unix, Git

CERTIFICATIONS

- 1) **NPTEL** - Python Programming, Fuzzy logic and system design - [link](#)
- 2) **Coursera Specialization** - Google IT Support, Deep Learning, Statistics with Python - [link](#)

ACHIEVEMENTS

- 1) **Award** : Best project and top performer in the 2022 batch of trainees at LTIMindtree.
- 2) **GATE Qualified** : GATE - CSE 2024 Qualified.